DETECT, DEFUSE, DEFEND: TACKLING TOXICITY WITH TRANSFORMERS

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Introduction

Online spaces thrive on respectful interaction, but in recent years, there has been a sharp rise in disruptive behavior that undermines digital communities. Toxicity in online communication refers to language that violates social norms—often subtly—causing emotional harm, escalating conflict, or discouraging open participation.

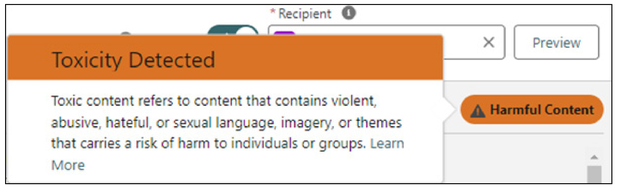

Figure 1: Systems like the one shown above attempt to automatically flag toxic content to maintain healthy discourse.

Platforms are increasingly concerned because toxic comments:

- Foster cyberbullying and harassment
- Undermine civil discourse
- Disproportionately silence vulnerable voices

Detecting toxicity requires models that understand context and tone, not just keywords. This project leverages transformer-based NLP models and strategic data approaches to classify toxic text with fairness and accuracy.

Preprocessing Techniques

- Retained only comment and toxic columns.
- Cleaned text: removed repetitions, timestamps, and special characters.

Handling Class Imbalance:

- Data Augmentation: Generated 20,000 toxic samples using the Vamsi/T5_Paraphrase_Paws model.
- Undersampling: Randomly reduced non-toxic samples to better match the toxic class.
- Final Distribution: 70,568 non-toxic vs. 35,294 toxic comments — a much fairer split.

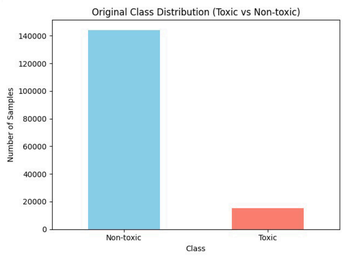

Figure 2: Given dataset

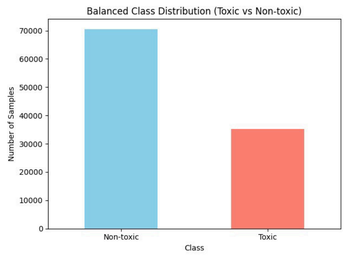

Figure 3: Balanced dataset

The original dataset exhibited significant class imbalance, with 144,277 non-toxic and only 15,294 toxic comments.

Baseline Toxicity Detection with Pretrained Transformers

To establish a reliable performance benchmark, we evaluated a publicly available transformer model — s-nlp/roberta_toxicity_classifier — on our balanced dataset of toxic and non-toxic comments.

About the Model:

- Built on the powerful roberta-large architecture.
- Fine-tuned on over 2 million comments labeled for toxicity from the Jigsaw dataset.
- Trained to generalize across a wide range of online comment types, including subtle and explicit toxic language.

Performance Highlights:

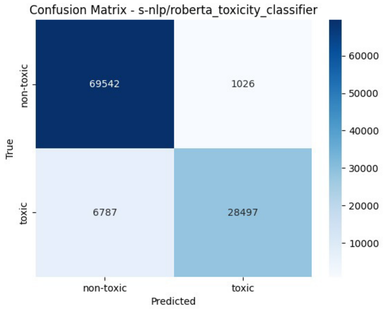

Figure 4: Confusion matrix for the pretrained RoBERTa model. The classifier achieves strong true positive and true negative rates.

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Classification Report:
              precision    recall  f1-score

   non-toxic       0.91      0.99      0.95
       toxic       0.97      0.81      0.88
```

Figure 5: Classification report. The model achieves 0.97 precision for toxic comments, though recall is slightly lower at 0.81.

This pre-trained model served as a robust baseline, allowing us to validate the quality and balance of our dataset before developing our own classification pipeline. Its strong generalization made it a suitable candidate for real-world toxicity detection scenarios.

Fine-Tuned google-bert/bert-base-uncased Model

To fine-tune both the bert-base-uncased and distilroberta-base models on the toxicity detection task, we employed a stratified 5-fold cross-validation setup to ensure balanced representation of the minority toxic class in each fold. The models were trained using the Hugging Face Trainer API with the following training configuration: a learning rate of 2e-5, batch size of 16 for both training and evaluation, and a total of 3 epochs. Weight decay regularization (0.01) was applied to prevent overfitting. Evaluation and model checkpointing were performed at the end of each epoch, and the best model for each fold was selected based on the F1 score for the toxic class (eval_f1_toxic).

We fine-tuned the `google-bert/bert-base-uncased` model using stratified 5-fold cross-validation on a rebalanced dataset (70k non-toxic, 35k toxic).

Performance Summary (Averaged Across Folds):

- F1–Score (Toxic): 0.984
- Precision (Toxic): 0.98
- Recall (Toxic): 0.99

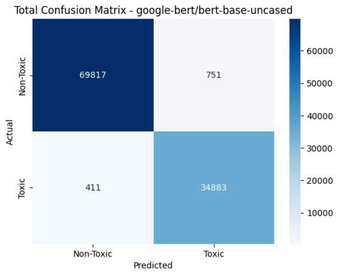

Figure 6: Confusion matrix of the fine-tuned model after rebalancing and training improvements. Shows improved true positive and true negative rates.

	Precision	Recall	F1-score
Non-Toxic	0.9941	0.9894	0.9917
Toxic	0.9789	0.9884	0.9836

Figure 7: Class distribution after data balancing and augmentation steps. The classes are well balanced for effective training.

Fine-tuned `distilbert/distilroberta-base`

Building on our success with the BERT-based model, we fine-tuned the lighter yet powerful distilbert/distilroberta-base architecture to evaluate whether a smaller model could deliver comparable performance while being faster and more resource-efficient.

Motivation: In real-world deployment scenarios, low-latency and efficiency matter. This experiment explored the trade-off between size and performance.

Performance Summary:

- F1–Score (Toxic): 0.962
- Precision (Toxic): 0.95
- Recall (Toxic): 0.97

Key Insight: While slightly lower than the full BERT model, the results are still strong — making it ideal for production environments with speed constraints.

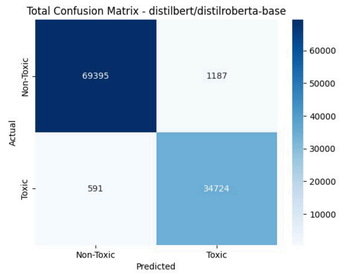

Figure 8: Confusion matrix for the fine-tuned DistilRoBERTa model. Shows reliable detection of both toxic and non-toxic comments.

	precision	recall	f1-score
Non-Toxic	0.998	0.975	0.986
Toxic	0.942	0.966	0.953

Figure 9: Classification report. Slight drop in precision compared to BERT, but significantly faster inference time

Comparisons

Based on Toxic F1 score and confusion matrix analysis, fine-tuned google-bert/bert-base-uncased outperformed both the pretrained RoBERTa model and fine-tuned distilbert/distilroberta-base. This makes it the most effective model for identifying minority toxic instances.

References

Data Varvara Logacheva, Daria Dementieva, Sergey Ustyantsev, et al. "ParaDetox: Detoxification with Parallel Data". In: Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL). 2022, pp. 6804–6818. URL: https://aclanthology.org/2022.acllong.469.

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